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PRACTICAL 1

Installation steps of Android Studio

① Download Android Studio

Visit https://developer.android.com/studio

Click "Download Android Studio" (choose your OS)

② Run the Installer

Windows : Run .exe file and follow prompts

macOS : Open .dmg and drag Android Studio to applications

Linux : Extract .zip and run studio.sh

③ Setup Wizard

Choose UI theme

Follow prompts to install SDK, emulator, etc

Click "Finish" to launch

④ Configure SDK

Accept licenses and install recommended components

Set up Android SDK location

⑤ Create a Project

Click "New Project" - Choose template

Configure project name, package and location

⑥ Run your App

Connect an Android device or use emulator

Click "Run"

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PRACTICAL 2(a)

Create an Android Application to display hello world run it on emulator as well as mobile

xml

```
<?xml version="1.0" encoding="utf-8"?>
<TextView
    xmlns:android="http://schemas.android.com/apk/res/
    android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:text="Hello World"
    android:textSize="24sp"
    android:gravity="center"/>
```

java

```
package com.example.helloworld;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

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PRACTICAL 2(b)

Android application to implement the use of various components of android studio.

xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/
    android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:id="@+id/txtTitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Android Components Eg"
        android:textSize="22sp"/>

    <EditView
        android:id="@+id/editName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter your name"/>

    <Button
        android:id="@+id/btnClick"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click Me"/>

    <ImageView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
    android:src="@mipmap/ic_launcher"/>
</LinearLayout>
```

java

```
package com.example.componentsapp;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    EditText editName;
    Button btnClick;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editName = findViewById(R.id.editName);
        btnClick = findViewById(R.id.btnClick);

        btnClick.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String name = editName.getText().toString();
                Toast.makeText(MainActivity.this, "Hello" +
                name, Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

PRACTICAL 3(a)

Create a basic login form using Android Studio

xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com
/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    android:gravity="center">

  <EditText
      android:id="@+id/e1"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Username"
      android:inputType="text"/>

  <EditText
      android:id="@+id/e2"
      android:layout_width="match_parent"
      android:layout_height="wrap_content
      android:hint="Password"
      android:inputType="textPassword"

  <Button
      android:id="@+id/btnLogin"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Login"

</LinearLayout>
```

java

```
package com.example.loginapp;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    EditText e1, e2;
    Button btnLogin;

    @Override
    protected void onCreate(Bundle savedInstanceState)
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        e1 = findViewById(R.id.e1);
        e2 = findViewById(R.id.e2);
        btnLogin = findViewById(R.id.btnLogin);

        btnLogin.setOnClickListener(new View.OnClickListener
() {
            @Override
            public void onClick(View v) {

                String username = e1.getText().toString();
                String password = e2.getText().toString();

                if (username.equals("admin") && password.
equals("1234")) {
                    Toast.makeText(MainActivity.this,
"Login Successful", Toast.LENGTH_SHORT).show(); }
                else {
```

```
                Toast.makeText(MainActivity.this, "Invalid
Username or Password", Toast.LENGTH_SHORT).show();
        }
    }
    });
  }
}
```

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PRACTICAL 3(b)

Android application implementing the if statement for basic authentication.

xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id/username"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username"/>

    <EditText
        android:id="@+id/password"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputText="textPassword"

    <Button
        android:id="@+id/loginbtn
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login"

</LinearLayout>
```

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java

```
package com.example.simpleauth;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    EditText username, password;
    Button loginbtn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        username = findViewById(R.id.username);
        password = findViewById(R.id.password);
        loginbtn = findViewById(R.id.loginbtn);

        loginbtn.setOnClickListener(new View.OnClickListener() {

        @Override
        public void onClick(View v) {

            String user = username.getText().toString();
            String pass = password.getText().toString();

            if (user.equals("admin") && pass.equals("123")) {
                Toast.makeText(MainActivity.this, "Login
Successful", Toast.LENGTH_SHORT).show(); }
            else {
                Toast.makeText(MainActivity.this, "Login
```

```
Failed", Toast.LENGTH_SHORT).show();
                }
            }
        });
    }
}
```

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PRACTICAL 4(a)

Create an android application to calculate age by taking DOB as an input.

xml

```
<?xml version="1.0" encoding="utf-8">
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <EditText
        android:id="@+id+e1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter DOB"
        android:inputType="number"/>

    <Button
        android:id="@+id/b1"
        android:layout_width="wrap_content"
        android:layout_width="wrap_content"
        android:text="Calculate Age"

    <TextView
        android:id="@+id/tResult"
        android:layout_width="wrap_content"
        android:layout_heigth="wrap_content"
        android:textSize="18sp"

</LinearLayout>
```

java

```
package com.example.agecalculator;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;
import java.util.Calendar;

public class MainActivity extends AppCompatActivity {

    EditText e1;
    Button b1;
    TextView tResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        e1 = findViewById(R.id.e1);
        b1 = findViewById(R.id.b1);
        tResult = findViewById(R.id.tResult);

        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                int birthYear = Integer.parseInt(e1.getText().
toString());
                int currentYear = Calendar.getInstance().getCalendar
.YEAR);
                int age = currentYear - birthYear;
                tResult.setText("Your Age is: " + age + " years");
            } });  } }
```

PRACTICAL 4(b)

Create an android application to calculate BMI by taking height and weight as input.

xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/eWeight"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter weight (kg)"
        android:inputType="numberDecimal"/>

    <EditText
        android:id="@+id/etHeight"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter height (cm)"
        android:inputType="numberDecimal"/>

    <Button
        android:id="@+id/b1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Calculate BMI"/>

    <TextView
        android:id="@+id/tResult"
```

```
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:textSize="18sp"

</LinearLayout>
```

java

```
package com.example.bmicalculator;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    EditText eWeight, eHeight;
    Button b1;
    TextView tResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(Bundle savedInstanceState);
        setContentView(R.layout.activity_main);

        eWeight = findViewById(R.id.eWeight);
        eHeight = findViewById(R.id.eHeight);
        b1 = findViewById(R.id.b1);
        tResult = findViewById(R.id.tResult);

        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                float weight = Float.parseFloat(eWeight.
getText.toString());
```

```
                float height =Float.parseFloat(eHeight.getText()
.toString())/100;
                float bmi = weight /(height* height);
                tResult.setText("BMI:" + bmi);
            }
        });
    }
}
```

PRACTICAL 5(a)

Android application to implement use of intent (Take only 2 intent)

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

  <Button
      android:id="@+id/btnNext"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Go to Second Activity"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.intentexample;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnNext;
```

```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnNext = findViewById(R.id.btnNext);

        btnNext.setOnClickListener(new View.OnClickListener()
{
            public void onClick(View v) {
                Intent intent = new Intent(MainActivity.this,
SecondActivity.class);
                startActivity(intent);
            }
        });
    }
}
```

activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/
apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to Second Activity"
        android:textSize="20sp"/>
</LinearLayout>
```

SecondActivity.java

```
package com.example.intentexample;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
public class SecondActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);
    }
}
```

AndroidManifest.xml

```
<activity android:name=".SecondActivity"/>
```

PRACTICAL 5(b)

Android application to implement multiple intents

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center">

    <Button
        android:id="@+id/btnSecond"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Open Second Activity"/>

    <Button
        android:id="@+id/btnThird"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content'
        android:text="Open Third Activity"/>
</LinearLayout>
```

activity_second.xml

```
<TextView xmlns:android="http://schemas.android.com/apk/res/android
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:text="This is Second Activity"
    android:gravity="center"
    android:textSize="20sp"/>
```

activity_third.xml

```
<TextView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:text="This is Third Activity"
    android:gravity="center"
    android:textSize="20sp"/>
```

MainActivity.java

```
package com.example.multipleintents;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnSecond, btnThird;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnSecond = findViewById(R.id.btnSecond);
        btnThird = findViewById(R.id.btnThird);

        btnSecond.setOnClickListener(new View.OnClickListener()
{
            public void onClick(View v) {
                Intent i = new Intent(MainActivity.this,
SecondActivity.class);
                startActivity(i); }
        });
```

```
        btnThird.setOnClickListener(new View.OnClickListener()
{
            public void onClick(View v) {
                Intent i = new Intent(MainActivity.this,
ThirdActivity.class);
                startActivity(i);
            }
        });
    }
}
```

SecondActivity.java

```
package com.example.multipleintents;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCopatActivity {

    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);
    }
}
```

ThirdActivity.java

```
package com.example.multipleintents;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class ThirdActivity extends AppsCompatActivity {
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_third);
    }
}
```

AndroidManifest.xml

```
<application>
    <activity android:name=".ThirdActivity"/>
    <activity android:name=".SecondActivity"/>
    <activity android:name=".MainActivity"/>
        <intent-filter>
            <action android:name="android.intent.action.
MAIN"/>
            <category android:name="android.intent.
category.LAUNCHER"/>
        </intent-filter>
    </activity>
</application>
```

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